Tree over all 1-Field Lagrangians
62,551 pts over 8 Lagrangians

See anything at IceCube-Gen2?
yes
See a signal 10x above the CTA (WW) limit?
yes
See anything at DARWIN?
yes
2925 pts, 273 regions, 5 lagrangians
LLM split
LLM-Agent Literature Search AMS-02 cosmic-ray positron spectrum hard e+ component: <sigma v>(e+e- equivalent) >= 3e-26 cm^3/s ? arXiv:2107.10261, arXiv:1306.3983 Status: Splits!
LLM-Agent Novel Experiment Proposal GC positron box-edge spectrometer (AMS-100/ALADInO class) e+ flux box lower edge E- >= 3 GeV ? Feasibility: unlikely
604 pts, 8 regions, 2 lagrangians
530 pts, 17 regions, 1 lagrangian
1540 pts, 1 region, 1 lagrangian
no
32078 pts, 4047 regions, 5 lagrangians
LLM split
LLM-Agent Literature Search AMS-02 cosmic-ray positron spectrum hard e+ component: <sigma v>(e+e- equivalent) >= 3e-26 cm^3/s ? arXiv:2107.10261, arXiv:1306.3983 Status: Splits!
LLM-Agent Novel Experiment Proposal GC positron box-edge spectrometer (AMS-100/ALADInO class) e+ flux box lower edge E- >= 1.5 GeV ? Feasibility: unlikely
883 pts, 73 regions, 3 lagrangians
8409 pts, 235 regions, 4 lagrangians
18528 pts, 1 region, 1 lagrangian
no
See anything at CTA (WW)?
yes
See anything at DARWIN?
yes
916 pts, 212 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search LHCb/CMS prompt A'->mumu scan narrow mumu peak: none | 1.0 | 2.2-2.5 | 2.5-3.0 | 12-15 | 19-25 | 37 GeV ? arXiv:1910.06926, arXiv:1912.04776 Status: Splits!
LLM-Agent Novel Experiment Proposal Belle II ISR gamma+mumu scan, 50/ab ISR mumu peak at 9.8-9.9 GeV (>=5 sigma) ? Feasibility: possible
4 pts, 1 region, 1 lagrangian
624 pts, 20 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal CTA GC spectral decomposition, Z'Z' cascade soft cascade component >= 3% of DM gamma flux ? Feasibility: unlikely
308 pts, 2 regions, 2 lagrangians
320 pts, 19 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal LHCb Upgrade-II peak signal strength sigma.BR(A'->mumu, 1.0 GeV) >= 1 pb ? Feasibility: possible
29 pts, 1 region, 1 lagrangian
4 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal LHCb Upgrade-II peak signal strength sigma.BR(A'->mumu, 2.3-2.5 GeV) >= 1 pb ? Feasibility: possible
6 pts, 1 region, 1 lagrangian
4 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
4 pts, 1 region, 1 lagrangian
41 pts, 1 region, 1 lagrangian
13 pts, 1 region, 1 lagrangian
no
1081 pts, 387 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search LHCb/CMS prompt A'->mumu scan narrow mumu peak: none | 1.0-2.2 | 6-9 | 17-24 GeV ? arXiv:1910.06926, arXiv:1912.04776 Status: Splits!
LLM-Agent Novel Experiment Proposal CTA deep-exposure GC spectral endpoint gamma spectrum endpoint >= 450 GeV ? Feasibility: possible
53 pts, 8 regions, 1 lagrangian
14 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal LHCb Upgrade-II peak signal strength sigma.BR(A'->mumu, 6-9 GeV) >= 0.5 pb ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
404 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Belle II ISR gamma+mumu scan, 50/ab ISR mumu peak at 9.8-9.9 GeV (>=5 sigma) ? Feasibility: possible
6 pts, 1 region, 1 lagrangian
229 pts, 9 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal Displaced-dimuon vertex spectrometer (LHCb-U2 class) displaced mumu peak (vertex 0.02-50 mm), 1-16 GeV: mass bin ? Feasibility: speculative
56 pts, 6 regions, 2 lagrangians
149 pts, 2 regions, 2 lagrangians
24 pts, 1 region, 1 lagrangian
no
See anything at DARWIN?
yes
See anything at CTA (bb)?
yes
4 pts, 4 regions, 1 lagrangian
no
53 pts, 47 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search LHCb inclusive dark-photon A'->mumu search mumu resonance with 1.2 <= m(mumu) <= 1.6 GeV seen ? arXiv:1603.08926, arXiv:1910.06926 Status: Splits!
3 pts, 1 region, 1 lagrangian
5 pts, 1 region, 1 lagrangian
no
See a signal 10x above the CTA (bb) limit?
yes
12 pts, 8 regions, 1 lagrangian
no
2733 pts, 643 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search Dark-photon resonance search (BaBar / LHCb / Belle II) ll resonance in 1-6 GeV with eps >= 3e-4 ? If seen: m(ll) < 1.05 GeV ? arXiv:1406.2980, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
LLM-Agent Novel Experiment Proposal IceCube-Gen2 nu-spectrum endpoint nu endpoint E_max < 400 GeV ? Feasibility: unlikely
10 pts, 1 region, 1 lagrangian
4 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Belle II / LHCb kinetic-mixing determination from A' yield sigma(e+e- -> gamma A')*BR(ll) => eps >= 5e-3 ? Feasibility: possible
104 pts, 8 regions, 1 lagrangian
280 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal LHCb Upgrade II inclusive A' -> mumu prompt mumu peak 1-6 GeV (eps >= 2e-5) ? displaced mumu vertex (ctau 0.1-30 mm) ? Feasibility: possible
790 pts, 12 regions, 2 lagrangians
393 pts, 7 regions, 2 lagrangians
527 pts, 7 regions, 2 lagrangians
no
BR(h->inv) >= 0.001?
yes
See anything at CTA (WW)?
yes
BR(h->inv) >= 0.0032?
yes
See anything at DARWIN?
yes
See a signal 10x above the CTA (WW) limit?
yes
24 pts, 24 regions, 3 lagrangians
no
280 pts, 273 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search SHiP displaced dark-photon search displaced l+l- vertices with m(ll) = 0.95-1.05 GeV, N > 0 ? arXiv:2011.05115, arXiv:2203.05090, arXiv:2005.01515 Status: Splits!
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal FCC-hh 100 TeV dilepton resonance scan ee/mumu resonance, 2 TeV < m(ll) < 3 TeV, sigma x BR >= 0.01 fb ? Feasibility: unlikely
4 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
See a signal 10x above the CTA (WW) limit?
yes
99 pts, 71 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search LHCb inclusive prompt dark-photon dimuon search A' -> mu mu resonance mass m(mumu) >= 20.5 GeV ? arXiv:1910.06926, arXiv:1603.08926 Status: Splits!
3 pts, 1 region, 1 lagrangian
27 pts, 1 region, 1 lagrangian
no
233 pts, 224 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search SHiP displaced dark-photon vertex search displaced ll/hadron vertex with m(pair) in 1.0-1.4 GeV ? arXiv:1504.04855, arXiv:2011.05115 Status: Splits!
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Short-baseline dump vertex spectrometer displaced mumu vertex 0.5-5 m from target, m(mumu) 1.4-2.0 GeV ? Feasibility: unlikely
3 pts, 1 region, 1 lagrangian
7 pts, 2 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal Ultimate prompt dimuon scouting prompt mumu resonance in 20-25 GeV with sigma x BR >= 3e-3 fb ? Feasibility: unlikely
4 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
See a signal 10x above the CTA (WW) limit?
yes
361 pts, 288 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search BaBar/Belle II/LHCb visible dark-photon search A' -> ll resonance, 1-17 GeV: epsilon >= 3e-4 ? arXiv:1406.2980, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
61 pts, 6 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal CTA Galactic-center gamma spectral-endpoint mass fit annihilation endpoint E_max: <83 / 83-87 / 87-92 / >92 GeV ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
15 pts, 3 regions, 1 lagrangian
no
2956 pts, 2780 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search B-factory + LHCb dark-photon resonance search A' -> ll peak, 1-70 GeV: epsilon >= 3e-4 ? arXiv:1406.2980, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
10 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Next-generation beam-dump displaced-vertex A' search displaced l+l- vertex, m_A' 1-10 GeV, epsilon >= 1e-6 ? Feasibility: unlikely
66 pts, 9 regions, 2 lagrangians
143 pts, 39 regions, 3 lagrangians
no
BR(h->inv) >= 0.0032?
yes
BR(h->inv) >= 0.01?
yes
See a signal 10x above the Fermi15yr (bb) limit?
yes
BR(h->inv) >= 0.032?
yes
See anything at DARWIN?
yes
BR(h->inv) >= 0.1?
yes
3 pts, 3 regions, 1 lagrangian
no
46 pts, 43 regions, 2 lagrangians
no
BR(h->inv) >= 0.1?
yes
1 pts, 1 region, 1 lagrangian
no
158 pts, 112 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search LHCb Run 3-6 inclusive dark-photon dimuon search prompt A'->mumu peak, 1.0-2.6 GeV, eps^2 >= 2e-10 ? arXiv:1603.08926, arXiv:1910.06926, arXiv:0711.4866 Status: Splits!
LLM-Agent Novel Experiment Proposal A' mass-lifetime mapping with LHCb vertexing m(A'->mumu) >= 1.35 GeV ? Feasibility: possible
5 pts, 1 region, 1 lagrangian
6 pts, 2 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Dwarf-spheroidal cascade gamma spectroscopy (Fermi-LAT/AMEGO-X) stacked-dwarf flux ratio F(1-5 GeV)/F(0.3-1 GeV) >= 0.35 ? Feasibility: possible
13 pts, 2 regions, 1 lagrangian
32 pts, 5 regions, 1 lagrangian
no
273 pts, 189 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search Fermi-LAT stacked dwarf-spheroidal spectral-endpoint (DM mass spectroscopy) gamma spectrum endpoint E_max >= 4.75 GeV ? arXiv:0711.4866, arXiv:1503.02641, arXiv:1503.01773 Status: Splits!
LLM-Agent Novel Experiment Proposal LHCb Upgrade II millimeter-displaced dimuon dark-photon search mu+mu- resonance 1.0-1.6 GeV, vertex 0.05-10 mm: seen ? Feasibility: unlikely
53 pts, 4 regions, 1 lagrangian
25 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Downstream long-lived dark-photon spectrometer mumu/pipi vertex 1-50 cm, mass 1.0-1.5 GeV: seen ? Feasibility: speculative
9 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
BR(h->inv) >= 0.032?
yes
See anything at CTA (bb)?
yes
See anything at DARWIN?
yes
37 pts, 37 regions, 2 lagrangians
no
6 pts, 6 regions, 1 lagrangian
no
See anything at DarkSide?
yes
BR(h->inv) >= 0.1?
yes
1 pts, 1 region, 1 lagrangian
no
11 pts, 6 regions, 1 lagrangian
no
BR(h->inv) >= 0.1?
yes
4 pts, 4 regions, 1 lagrangian
no
See anything at DARWIN?
yes
3 pts, 3 regions, 3 lagrangians
no
75 pts, 71 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search Planck CMB exotic energy-injection bound p_ann = f_eff<sigma v>/m_DM >= 3.2e-28 cm^3 s^-1 GeV^-1 ? arXiv:1807.06209, arXiv:1506.03811, arXiv:0711.4866 Status: Splits!
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
See anything at CTA (bb)?
yes
See anything at DARWIN?
yes
2 pts, 2 regions, 2 lagrangians
no
27 pts, 27 regions, 2 lagrangians
no
See anything at DarkSide?
yes
4 pts, 4 regions, 1 lagrangian
no
See anything at DARWIN?
yes
1 pts, 1 region, 1 lagrangian
no
207 pts, 142 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search Belle II gamma + dark-photon search A' at eps >= 1e-3, m(A') 1-8 GeV: mumu peak / invisible peak / nothing? arXiv:1406.2980, arXiv:1702.03327, arXiv:1808.10567 Status: Splits!
LLM-Agent Novel Experiment Proposal A' dimuon line spectroscopy m(mumu) peak: 1.00 / 1.25 / 1.33 GeV (+-0.02 GeV)? Feasibility: possible
20 pts, 2 regions, 2 lagrangians
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal A' signal-rate mixing fit sigma(ee -> gamma A') x BR(mumu) implies eps >= 0.05? Feasibility: possible
4 pts, 1 region, 1 lagrangian
16 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Z-lineshape EW fit, kinetic mixing EW-fit eps >= 0.03 at m(Z') ~ 21 GeV? Feasibility: possible
3 pts, 1 region, 1 lagrangian
46 pts, 7 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal SHiP beam-dump displaced A' >= 3 displaced ee/mumu vertices with m(ll) ~ 1.0 GeV? Feasibility: possible
9 pts, 3 regions, 2 lagrangians
37 pts, 4 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Halo-ensemble DM self-interaction sigma_T/m: >= 0.015 / 0.003-0.015 / < 0.003 cm^2/g? Feasibility: speculative
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Belle II 50/ab displaced-vertex A' displaced ll vertex peak at m = 1.25 GeV? Feasibility: unlikely
3 pts, 1 region, 1 lagrangian
34 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Ultra-faint SIDM tier sigma_T/m: >= 5e-4 / 1e-6-5e-4 / < 1e-6 cm^2/g? Feasibility: speculative
28 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
See a signal 10x above the Fermi15yr (bb) limit?
yes
811 pts, 370 regions, 3 lagrangians
LLM split
LLM-Agent Literature Search LHCb/Belle II prompt dark-photon search A' -> mu mu peak at 1-3.3 GeV with epsilon >= 3e-4 ? arXiv:1710.02867, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
LLM-Agent Novel Experiment Proposal Dark-photon mass spectroscopy (LHCb dimuon) m(A' -> mu mu) >= 2 GeV ? Feasibility: possible
38 pts, 7 regions, 1 lagrangian
16 pts, 2 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal GC/dwarf annihilation spectral-endpoint fit (Fermi-LAT) gamma spectrum cutoff >= 3.6 GeV ? Feasibility: possible
403 pts, 19 regions, 1 lagrangian
8 pts, 1 region, 1 lagrangian
no
See anything at CTA (bb)?
yes
See anything at DARWIN?
yes
See a signal 10x above the CTA (bb) limit?
yes
68 pts, 62 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search AMS-02 positron spectrum: sharp box edge from Z' cascade e+ flux cutoff edge at E >= 65 GeV ? arXiv:1306.3983, arXiv:1309.2570, arXiv:1503.01773 Status: Splits!
4 pts, 1 region, 1 lagrangian
4 pts, 1 region, 1 lagrangian
no
3 pts, 3 regions, 2 lagrangians
no
See a signal 10x above the CTA (bb) limit?
yes
70 pts, 66 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search LHC low-mass prompt dimuon dark-photon search (LHCb inclusive + CMS scouting) narrow mumu resonance, 10.5 < m_mumu < 17 GeV, eps >= 5e-3 ? arXiv:1910.06926, arXiv:1912.04776, arXiv:1801.04847 Status: Splits!
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
38 pts, 36 regions, 2 lagrangians
no
See anything at DarkSide?
yes
8 pts, 6 regions, 1 lagrangian
no
See anything at DARWIN?
yes
2 pts, 2 regions, 2 lagrangians
no
190 pts, 172 regions, 5 lagrangians
LLM split
LLM-Agent Literature Search B-factory dark-photon search (visible + mono-photon) e+e- -> gamma A' signal, 1-5 GeV, eps >= 1e-3 ? arXiv:1406.2980, arXiv:1702.03327, arXiv:1910.06926 Status: Splits!
LLM-Agent Novel Experiment Proposal A' mass spectroscopy at Belle II / LHCb m(A') < 3 GeV, 3-4.5 GeV, or > 4.5 GeV ? Feasibility: possible
6 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal LHCb Upgrade II / full-lumi Belle II prompt dimuon prompt mumu resonance 0.95-1.1 GeV, eps >= 5e-5 ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
9 pts, 2 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal Low-threshold Si/Ge recoil-spectrum endpoint nuclear recoils at E_R >= 2 keV, sigma_SI ~ 1e-44 cm2 ? Feasibility: unlikely
3 pts, 1 region, 1 lagrangian
6 pts, 1 region, 1 lagrangian
no
See anything at CTA (bb)?
yes
See a signal 10x above the CTA (bb) limit?
yes
See anything at DARWIN?
yes
26 pts, 26 regions, 2 lagrangians
no
912 pts, 600 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search HL-LHCb dark-photon A'->mumu search A'->mumu bump, 1-30 GeV, eps >= 1e-4: m_mumu >= 10 GeV / < 10 GeV / unseen ? arXiv:1910.06926, arXiv:1603.08926, arXiv:1808.08865 Status: Splits!
LLM-Agent Novel Experiment Proposal A' signal-strength mixing determination eps from A' yield: >= 1e-2 / 3e-4 - 1e-2 / < 3e-4 ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
9 pts, 2 regions, 1 lagrangian
4 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal A' signal-strength mixing determination (low mass) eps from A' yield >= 3e-3 ? Feasibility: possible
118 pts, 8 regions, 1 lagrangian
21 pts, 3 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal CTA+Fermi GC spectral endpoint gamma endpoint E_end: < 20 GeV / 20-58 GeV / >= 58 GeV ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
7 pts, 2 regions, 2 lagrangians
177 pts, 20 regions, 3 lagrangians
LLM-Agent Novel Experiment Proposal Ultra-deep displaced dimuon A' search A'->mumu signal at eps >= 2e-5, 1-20 GeV ? Feasibility: unlikely
30 pts, 6 regions, 2 lagrangians
147 pts, 14 regions, 3 lagrangians
LLM-Agent Novel Experiment Proposal Percent-level endpoint energy calibration E_end >= 66 GeV ? Feasibility: unlikely
144 pts, 13 regions, 2 lagrangians
3 pts, 1 region, 1 lagrangian
no
See a signal 10x above the Fermi15yr (bb) limit?
yes
389 pts, 226 regions, 2 lagrangians
LLM split
LLM-Agent Literature Search Fermi-LAT gamma-ray endpoint of the detected annihilation signal E_cut >= 3.8 GeV ? arXiv:0711.4866, arXiv:1503.01773, arXiv:1503.02641 Status: Splits!
LLM-Agent Novel Experiment Proposal Inclusive prompt A'->mumu scan at LHCb Upgrade II eps at m_mumu 1.0-1.6 GeV: >=2.5e-4 / >=2e-5 / none ? Feasibility: possible
3 pts, 1 region, 1 lagrangian
22 pts, 2 regions, 1 lagrangian
128 pts, 6 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Belle II gamma + mumu dark-photon scan near 1 GeV eps: >=1e-2 / >=3e-5 / none ? Feasibility: possible
8 pts, 2 regions, 1 lagrangian
3 pts, 1 region, 1 lagrangian
6 pts, 2 regions, 1 lagrangian
no
52 pts, 52 regions, 2 lagrangians
no
See a signal 10x above the Fermi15yr (bb) limit?
yes
1 pts, 1 region, 1 lagrangian
no
1217 pts, 443 regions, 5 lagrangians
LLM split
LLM-Agent Literature Search Belle II + LHCb visible dark-photon search prompt l+l- resonance, 0.2-70 GeV, eps >= 3e-4? measure m(ll), sigma*BR arXiv:1406.2980, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
LLM-Agent Novel Experiment Proposal Dwarf-galaxy DM self-interaction census sigma_self/m >= 1e-5 cm^2/g? Feasibility: speculative
5 pts, 1 region, 1 lagrangian
75 pts, 3 regions, 1 lagrangian
36 pts, 6 regions, 2 lagrangians
3 pts, 1 region, 1 lagrangian
8 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal LHCb Upgrade II inclusive displaced A'->mumu search mumu vertex displaced > 3 mm, m(mumu) 0.7-2 GeV? Feasibility: possible
87 pts, 16 regions, 2 lagrangians
575 pts, 11 regions, 3 lagrangians
LLM-Agent Novel Experiment Proposal FCC-ee Tera-Z radiative-return dark-photon scan gamma + narrow mumu/invisible recoil peak? prompt, displaced >0.3 mm, or none Feasibility: unlikely
7 pts, 1 region, 1 lagrangian
18 pts, 3 regions, 2 lagrangians
87 pts, 6 regions, 2 lagrangians
463 pts, 1 region, 1 lagrangian
no
See a signal 10x above the CTA (WW) limit?
yes
See anything at DARWIN?
yes
330 pts, 259 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search EW precision fit for gamma-Z' kinetic mixing (LEP + Z-pole programs) EW-fit kinetic mixing epsilon >= 0.03 at m_Z' ~ 15 GeV ? arXiv:1006.0973, arXiv:1412.0018 Status: Splits!
LLM-Agent Novel Experiment Proposal HL-LHC soft displaced-dimuon dark-cascade endpoint m(mumu) endpoint >= 1.5 GeV ? Feasibility: unlikely
69 pts, 12 regions, 1 lagrangian
8 pts, 2 regions, 1 lagrangian
LLM-Agent Novel Experiment Proposal Tera-Z radiative-return A' scan m(mumu) peak < 14 GeV ? sigma.BR(ee->gamma A'->gamma mumu) >= 1 fb ? Feasibility: unlikely
3 pts, 1 region, 1 lagrangian
5 pts, 1 region, 1 lagrangian
3 pts, 1 region, 1 lagrangian
no
See anything at Fermi15yr (WW)?
yes
See a signal 32x above the CTA (WW) limit?
yes
10 pts, 6 regions, 2 lagrangians
no
4 pts, 4 regions, 1 lagrangian
no
5543 pts, 2901 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search AMS-02 cosmic-ray antiproton spectrum pbar/p excess over secondary background, 5-50 GeV: > 10% ? arXiv:1610.03071, arXiv:1903.02549, arXiv:1511.08787 Status: Splits!
LLM-Agent Novel Experiment Proposal CTA deep-GC endpoint spectroscopy gamma spectrum cutoff energy >= 200 GeV ? Feasibility: possible
412 pts, 76 regions, 4 lagrangians
1743 pts, 73 regions, 4 lagrangians
LLM-Agent Novel Experiment Proposal Next-gen magnetic-spectrometer positron edge e+ flux: sharp spectral edge at E >= 180 GeV ? Feasibility: unlikely
373 pts, 11 regions, 2 lagrangians
36 pts, 9 regions, 3 lagrangians
no
See anything at CTA (WW)?
yes
See anything at DARWIN?
yes
29 pts, 29 regions, 2 lagrangians
no
7161 pts, 6421 regions, 5 lagrangians
LLM split
LLM-Agent Literature Search LHCb + Belle II visible dark-photon search A'->l+l- resonance, 0.2 < m < 70 GeV, eps >= 3e-4 ? arXiv:1910.06926, arXiv:1808.10567, arXiv:2005.01515 Status: Splits!
39 pts, 10 regions, 2 lagrangians
LLM-Agent Novel Experiment Proposal CTA Galactic-centre spectral-endpoint fit annihilation spectrum endpoint E_cut >= 130 GeV ? Feasibility: possible
66 pts, 7 regions, 2 lagrangians
791 pts, 159 regions, 3 lagrangians
LLM-Agent Novel Experiment Proposal Near-dump displaced Z' spectrometer displaced l+l- vertex, mass 0.3-8 GeV, eps >= 1e-6 ? Feasibility: unlikely
147 pts, 21 regions, 2 lagrangians
710 pts, 145 regions, 3 lagrangians
LLM-Agent Novel Experiment Proposal Tera-Z electroweak Z-Z' mixing fit |Delta sin^2 theta_eff^lept| >= 1e-5 ? Feasibility: unlikely
34 pts, 5 regions, 2 lagrangians
823 pts, 161 regions, 3 lagrangians
no
See anything at DARWIN?
yes
7 pts, 7 regions, 3 lagrangians
no
1066 pts, 159 regions, 4 lagrangians
LLM split
LLM-Agent Literature Search Dark-photon dimuon scan (BaBar/Belle II/LHCb) e+e- -> gamma A'(->mumu) resonance sigma > 1 fb; m_A' >= 1.1 GeV ? arXiv:1406.2980, arXiv:1910.06926, arXiv:1808.10567 Status: Splits!
LLM-Agent Novel Experiment Proposal Extreme-precision halo self-interaction survey sigma_self/m in halos >= 5.3e-7 + 1e-12 cm^2/g ? Feasibility: speculative
469 pts, 4 regions, 1 lagrangian
17 pts, 3 regions, 1 lagrangian
7 pts, 1 region, 1 lagrangian
LLM-Agent Novel Experiment Proposal Near-target displaced-vertex proton-dump spectrometer displaced l+l-/pipi vertex 0.1-3 m from dump, m(ll) 1.0-2.7 GeV: >= 10 events ? Feasibility: unlikely
7 pts, 1 region, 1 lagrangian
417 pts, 3 regions, 2 lagrangians